Consistent Modified Gravity Weak Lensing in Energy-Flow Cosmology:

The μ–Σ Eigenmode, Destructive Interference, and E_G as the Orthogonal Probe

Research Note — EFC White Paper Series

Morten Magnusson[1]

[1]Symbiose Research

ORCID: 0009-0002-4860-5095

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Abstract. We derive the observable structure of Energy-Flow Cosmology (EFC) in weak lensing, moving from the phenomenological Case A ($P_{\text{eff}} = P_{\Lambda\text{CDM}} \times \Sigma^2$) to the consistent Case B where $\mu(k,z)$ enters the growth equations and $\Sigma(k,z)$ enters the lensing kernel via modified Poisson equations. The EFC bridge maps field parameters $(K_0, m^2) = (1.66, 0.0035)$ to $(\mu_0, \Sigma_0) = (0.94, 1.05)$ at the calibration anchor $k = 0.05$ h/Mpc, with five independent validation checks passing against Kill-Test v6 (DOI: 10.6084/m9.figshare.31368433). We show analytically that weak lensing constrains only a single effective parameter $\eta_{\text{eff}} = \gamma(\mu-1)+(\Sigma-1) = 0.017$, with the orthogonal direction η_{deg} invisible to shear data. Growth suppression from $\mu < 1$ cancels 66% of the lensing amplification from $\Sigma > 1$, producing destructive interference. The expected signal is $\Delta\chi^2 \in [-2, +2]$ against KiDS-1000. We prove that the E_G statistic (galaxy-galaxy lensing divided by redshift-space distortions) measures Σ/μ^γ — the *ratio*, not the product — and is therefore orthogonal to weak lensing. In E_G, μ and Σ interfere constructively, yielding $E_G^{\text{EFC}}/E_G^{\text{GR}} = 1.086$ (+8.6%). Combined, weak lensing and E_G fully separate μ and Σ: $\Sigma = \sqrt{AB}$, $\mu^\gamma = \sqrt{A/B}$. Five sealed numerical predictions are registered prior to any Case B run.

1 Introduction

Energy-Flow Cosmology (EFC) modifies gravity at cosmological scales through an entropic flow mechanism that introduces scale-dependent modifications to the Poisson equations [1]. At the perturbation level, this is parametrised by two functions: $\mu(k,z)$, which modifies Newton's coupling and affects structure growth, and $\Sigma(k,z)$, which modifies the lensing potential.

A previous analysis (Case A, DOI: 10.6084/m9.figshare.31271917 [2]) applied a phenomenological lensing response $P_{\text{eff}} = P_{\Lambda\text{CDM}} \times \Sigma^2(k,z)$ to KiDS-1000 cosmic shear, yielding $\Delta(-2\ln\mathcal{L}) = -50.9$. However, Case A applies a phenomenological rescaling without growth feedback — the power spectrum $P(k)$ is computed from standard growth equations, then multiplied by Σ^2. The reduced chi-squared $\chi^2_{\text{red}} > 5.0$ for the shear module suggests that the fit improvement reflects this inconsistency.

This note presents Case B: consistent modified gravity where μ enters the growth ordinary differential equation and Σ enters the lensing kernel simultaneously. We derive the observable eigenstructure of this system, identify the source of signal cancellation, and prove which additional observables break the resulting degeneracy.

2 The EFC Bridge

2.1 Modified Poisson Equations

In modified gravity, the Bardeen potentials Φ and Ψ satisfy:

$$k^2\Phi = -4\pi G a^2 \mu(k,z)\,\bar{\rho}\,\delta \tag{1}$$

$$k^2(\Phi + \Psi) = -8\pi G a^2 \Sigma(k,z)\,\bar{\rho}\,\delta \tag{2}$$

with gravitational slip $\eta \equiv \Psi/\Phi = 2\Sigma/\mu - 1$.

2.2 Bridge Equations

The EFC modification strength is:

$$R(k,a) = K_0 \cdot \Theta\left(\frac{\rho_m(a)}{\rho_*}\right) \cdot (\Gamma'\dot{\phi})^2 \cdot \frac{a^4}{k^4} \tag{3}$$

where $\Theta(\rho/\rho_*) = \exp[-(\rho/\rho_*)^2]$ is the density gate ensuring GR recovery in high-density environments. The bridge functions are:

$$\mu(k) = \frac{1}{1 + R(k)} \tag{4}$$

$$\eta(k) = 1 + \frac{(\eta_{\mathrm{ref}} - 1)}{R_{\mathrm{ref}}} \cdot R(k), \quad \text{capped at 2} \tag{5}$$

$$\Sigma(k) = \tfrac{1}{2}(1 + \eta) \cdot \mu \tag{6}$$

2.3 Validation

The bridge reproduces the calibration anchor from Kill-Test v6 [1] with five independent checks:

Table 1: Bridge validation against Kill-Test v6 calibration values.

Scale	k [h/Mpc]	μ	Σ	Status
CMB lensing (anchor)	0.050	0.940	1.048	✓
BAO / LSS	0.100	0.996	1.003	✓
Galactic	5.000	1.000	1.000	✓
Solar System	50.00	1.000	1.000	✓
Slip at anchor	0.050	$\eta = 1.230$	—	✓

GR is recovered at $k \gtrsim 0.5$ h/Mpc via two independent mechanisms: the k^{-4} suppression of $R(k)$ and the density gate $\Theta \to 0$ at galactic and solar-system densities.

3 The Observable Eigenstructure

3.1 Weak Lensing Power Spectrum

The cosmic shear angular power spectrum between tomographic bins i and j is:

$$C_\ell^{ij} \propto \int \frac{\Sigma^2(k,z) \cdot D^2(\mu;z)}{\chi^2(z)} P_0(k)\, W_i(z)\, W_j(z)\, dz \tag{7}$$

where $D(\mu) \approx D_{\mathrm{GR}} \cdot \mu^\gamma$ with $\gamma \approx 0.55$ the growth index [3]. This approximation is valid for small deviations from GR ($|\mu - 1| \lesssim 0.1$) at low redshift ($z \lesssim 1$), which is satisfied at the EFC anchor. The ratio to GR is:

$$\frac{C_\ell^{\mathrm{EFC}}}{C_\ell^{\mathrm{GR}}} = \Sigma^2 \cdot \mu^{2\gamma} \tag{8}$$

3.2 The Effective Parameter

Linearising Eq. (8) around GR ($\mu = 1$, $\Sigma = 1$). Writing $\Sigma = 1 + \delta_\Sigma$ and $\mu = 1 + \delta_\mu$ with $|\delta_\Sigma|, |\delta_\mu| \ll 1$:

$$\Sigma^2 \mu^{2\gamma} \approx (1 + 2\delta_\Sigma)(1 + 2\gamma\delta_\mu) \approx 1 + 2[\delta_\Sigma + \gamma\delta_\mu] \tag{9}$$

Defining $\eta_{\mathrm{eff}} \equiv \gamma(\mu - 1) + (\Sigma - 1)$:

$$\boxed{\frac{C_\ell^{\mathrm{EFC}}}{C_\ell^{\mathrm{GR}}} \approx 1 + 2\eta_{\mathrm{eff}}, \qquad \eta_{\mathrm{eff}} \equiv \gamma(\mu - 1) + (\Sigma - 1)} \tag{10}$$

This is the *only* combination weak lensing constrains. The orthogonal direction,

$$\eta_{\mathrm{deg}} \equiv (\Sigma - 1) - \gamma(\mu - 1) \tag{11}$$

lies along the isocontour $C_\ell^{\mathrm{EFC}} = C_\ell^{\mathrm{GR}}$. Setting $\eta_{\mathrm{eff}} = 0$ gives $(\Sigma - 1) = -\gamma(\mu - 1)$, which to leading order yields $\Sigma \approx \mu^{-\gamma}$: the degeneracy line. This direction is invisible to shear data.

3.3 Signal Decomposition

Table 2: Decomposition of the effective parameter at the EFC anchor.

Component	Formula	Value	Interpretation
Σ boost	$\Sigma_0 - 1$	$+0.050$	lensing amplification
μ drag	$\gamma(\mu_0 - 1)$	-0.033	growth suppression
Net η_{eff}	sum	$\mathbf{+0.017}$	**66% cancelled**

The cancellation ratio is $|\gamma(\mu_0 - 1)|/|(\Sigma_0 - 1)| = 0.033/0.050 = 0.66$. Only 34% of the lensing signal survives as an observable effect. This explains quantitatively why Case A (which bypassed growth suppression) produced $\Delta(-2\ln\mathcal{L}) = -50.9$ while Case B is expected to yield $\Delta\chi^2 \approx 0$: Case A measured Σ alone; Case B measures η_{eff}, in which μ partially cancels Σ.

4 Scale and Redshift Dependence

The EFC modification is localised in both scale and redshift:

Table 3: Scale dependence of the net C_ℓ ratio at $z_{\rm eff} = 0.3$.

ℓ	k [h/Mpc]	$\mu(k)$	$\Sigma(k)$	$C_\ell^{\rm EFC}/C_\ell^{\rm GR}$
50	0.056	0.960	1.032	1.019
100	0.111	0.997	1.002	1.001
200	0.222	1.000	1.000	1.000
500	0.556	1.000	1.000	1.000

The signal is concentrated at $\ell \lesssim 100$ (angular scales $\theta \gtrsim 1°$) and vanishes above $\ell \approx 200$ due to the k^{-4} suppression in Eq. (3). In redshift, the effect peaks in low-z tomographic bin pairs — $(1,1)$, $(1,2)$, $(2,2)$ — where the scale factor $a \to 1$ maximises $R(k,a) \propto a^4$.

Observational consequence: $\xi_+(\theta)$ at $\theta > 1°$ is the only shear observable with significant EFC content. $\xi_-(\theta)$, which is dominated by small scales, shows no EFC effect.

5 Breaking the Degeneracy: E_G

5.1 The Orthogonal Observable

The E_G statistic [4] combines galaxy–galaxy lensing ($\propto \nabla^2(\Phi + \Psi)$) with redshift-space distortions ($\propto f\,\delta$, where $f = d\ln D/d\ln a$). In GR, $E_G = \Omega_m/f$. In modified gravity, the lensing numerator picks up Σ while the RSD denominator picks up μ^γ through f, giving:

$$E_G \propto \frac{\Sigma}{\mu^\gamma} \tag{12}$$

This is the *ratio* of lensing to growth — orthogonal to the weak lensing eigenmode (the product). Define:

$$A \equiv \Sigma \cdot \mu^\gamma = 1.015 \quad \text{(WL eigenmode, product)} \tag{13}$$

$$B \equiv \Sigma/\mu^\gamma = 1.086 \quad (E_G \text{ eigenmode, ratio}) \tag{14}$$

Combined:

$$\boxed{\Sigma = \sqrt{A \cdot B}, \qquad \mu^\gamma = \sqrt{A/B}} \tag{15}$$

This provides full algebraic separation of μ and Σ from two observables.

5.2 Constructive vs. Destructive Interference

Table 4: Interference structure across observables. In E_G, μ and Σ reinforce rather than cancel.

Observable	μ-dep.	Σ-dep.	EFC signal	Interference	Breaks degen.?
WL C_ℓ	$\mu^{2\gamma}$	Σ^2	$+1.5\%$	destructive	No
E_G	$\mu^{-\gamma}$	Σ	$+8.6\%$	constructive	Yes
RSD $f\sigma_8$	μ^γ	none	-3.4%	—	Yes (pure μ)
ISW voids	$(1+\eta)\mu$	indirect	$+10\%$	—	Yes (unique)

The physical reason: in weak lensing, $\mu < 1$ suppresses the amplitude of $P(k)$ while $\Sigma > 1$ amplifies the lensing projection. These oppose each other. In E_G, $\Sigma > 1$ increases the numerator while $\mu^\gamma < 1$ decreases the denominator. Both shifts push E_G upward.

5.3 Data Availability

E_G measurements from KiDS-1000 $\times$ BOSS DR12 are already published [5]. No new observations are required — only a re-analysis with the EFC bridge.

6 Sealed Pre-Run Predictions

The following five predictions are registered prior to any Case B run. They are falsifiable by the Phase 1 minimisation:

1. $\eta_{\text{eff}} = 0.017$ at the EFC anchor (Eq. 10).
2. $\text{corr}(\mu_0, \Sigma_0) \in [-0.95, -0.80]$, expected from the Fisher eigenvector of the $\Sigma = \mu^{-\gamma}$ degeneracy.
3. $\Delta\chi^2(\text{Case B, WL only}) \in [-2, +2]$.
4. $E_G^{\text{EFC}}/E_G^{\text{GR}} = 1.086$ (Eq. 14; future test).
5. Cancellation ratio $= 0.66$ (model-independent given $\gamma \approx 0.55$).

Prediction (2) is the primary diagnostic: if the Phase 1 posterior shows $\text{corr}(\mu_0, \Sigma_0)$ outside $[-0.95, -0.80]$, the analytical eigenstructure derived here requires revision.

7 Discussion

7.1 Why Case A Succeeded and Case B Will Not

Case A applied Σ^2 as a post-hoc rescaling, bypassing the growth equation entirely. This is equivalent to setting $\mu = 1$ while allowing $\Sigma \neq 1$ — breaking the physical link between the two potentials. The $\Delta(-2\ln\mathcal{L}) = -50.9$ reflected the full Σ boost without the μ drag. Case B restores the link; the drag cancels 66% of the boost.

7.2 The Self-Cancellation as a Feature

The destructive interference is not a weakness of EFC. It is a structural consequence of any modification that couples μ and Σ through a common field $R(k, a)$. The result is a sharp, falsifiable prediction: EFC's lensing signal is small in shear but large in E_G. Any future lensing analysis that finds a large signal *without* a corresponding E_G signal would rule out the EFC bridge structure.

7.3 Limitations

The growth-index approximation $D \propto \mu^\gamma$ with $\gamma = 0.55$ is accurate to $\sim 1\%$ for $|\mu - 1| < 0.1$ [3]. At the EFC anchor ($\mu = 0.94$), the linearised ratio $1 + 2\eta_{\rm eff} = 1.034$ agrees with the exact expression $\Sigma^2\mu^{2\gamma} = 1.030$ to within 0.4%. The scale dependence uses the Limber approximation; sub-percent corrections from beyond-Limber terms are neglected.

7.4 Relation to Other EFC Hull Closures

This work addresses Hull #4 (Case B weak lensing) from the EFC validation roadmap. The eigenmode analysis directly informs Hull #3 (nested sampling) by identifying the optimal parameter basis, and Hull #6 (ISW void stacking) by showing that ISW probes a unique combination $(1+\eta)\mu$ not accessible to either WL or E_G.

Data and Code Availability

The bridge implementation (`efc_case_b_bridge.py`) with self-validation, the cobaya configuration (`efc_caseB_kids1000.yaml`), and the pre-run prediction files (`case_b_prediction.json`, `degeneracy_analysis.json`) are available in the Symbiose Research repository and will be deposited with this note.

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References

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